Shannon Crandley

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Education

Tufts University Expected May 2027

Bachelor of Science in Computer Engineering (GPA: 3.47 / 4.00) *Medford, MA*

Relevant Coursework: Embedded Systems, Computer Organization, Computer Communication Networks, Advanced Computer Architecture, Digital Signal Processing, Linear Systems, Parallel Computing, Data Structures

Technical Skills

Languages: C, C++, Python, Bash, VHDL, TypeScript, Java
Embedded & Hardware: STM32, ESP32, Arduino, FPGA, ADCs, GPIO, PWM, I²C, SPI, UART, ARM, RISC-V, x86
Tools: Git, Linux, Jira, MATLAB, LTspice, Docker, Nix
Software: Node.js, Svelte/SvelteKit, PostgreSQL
Fabrication: Soldering, CAD, 3D Printing, Laser Cutting

Experience

Mojotech May 2026 – Aug 2026

Software Engineer Intern *Providence, RI*

- Engineered a TypeScript application for internal use, integrating Jira and GitHub APIs to automate issue enhancement using agentic AI and improve project tracking workflows
- Participated in code reviews and collaborated with engineers to improve code quality, maintainability, and adherence to team standards
- Presented application architecture, implementation decisions, and project impact to company-wide engineering and technical teams

Tufts University Sept 2025 – Dec 2025

Research Assistant — MemSysExplorer *Medford, MA*

- Developed Python and Bash automation scripts to collect performance data from CPU and GPU memory-system benchmarks
- Built analysis pipelines to identify performance trends across different memory architectures
- Generated datasets used for research focused on memory behavior and system performance

Tufts University Sept 2026 – Present

Teaching Assistant - Junior Design (Robotics) *Medford, MA*

- Taught workshops on WebSockets, Git, CAD, and soldering
- Guided students through robot development, troubleshooting embedded systems, and integrating hardware and software

Projects

Voice Synthesizer | *STM32L432KC Microcontroller, C, ADC, Microphone, LCD*

- Developed an embedded C application on an STM32L432 microcontroller for real-time pitch synthesis using microphone input sampled through ADC channels
- Implemented pitch control and user interaction through LCD display and physical button inputs

Self-Navigating Robotic Vehicle | *Arduino Uno WiFi Rev2, C, Sensors, Motor Control*

- Designed sensor systems using phototransistors and IR sensors for color and distance detection
- Developed embedded C control logic integrating sensor data
- Implemented motor control with an H-bridge driver and coordinated movement through embedded C firmware
- Developed communication between robots via a server to coordinate movement and stopping behavior

Pac-Man FPGA System | *FPGA, VHDL, NES Controller Interface, VGA Display*

- Implemented Pac-Man game logic in VHDL on an FPGA, generating real-time graphics through VGA output
- Integrated an NES controller interface with FPGA-based game logic for real-time player input
- Designed digital modules for player movement, collision detection, and video rendering